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Open Targets: 10 years of partnership in target discovery

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The Open Targets consortium was founded in 2014 to translate insights from genetics and functional genomics into identifying and prioritizing therapeutic targets. We highlight key achievements and insights stemming from this partnership of academic and industry scientists.

Introduction

Many candidate medicines fail during clinical trials, when the safety and efficacy of the therapeutic approach is tested. These failures increase the cost of bringing new treatments to patients. They stem from our incomplete knowledge of how drug targets are linked to disease and our limited ability to predict the effect of modulating a target on disease.

As one potential solution to this problem, studies have indicated that drug targets supported by genetic evidence have a higher probability of success in drug development[1]. The Open Targets consortium was founded in 2014 to harness insights from genetics and functional genomics and transform target identification and prioritization. Industry and academic scientists work together to shape the Open Targets research programme, which has awarded 95 projects to date. The consortium comprises the founding partners — the Wellcome Sanger Institute, EMBL's European Bioinformatics Institute (EMBL-EBI) and the pharmaceutical company GSK — and more recent members, the companies Genentech, MSD, Pfizer, and Sanofi. Throughout its history, the companies Biogen, Takeda and Celgene (now BMS) have also counted amongst the Open Targets partners. Consortium findings are shared within the partnership before publication, then contributed to the public domain.

Establishing the Open Targets research programme

Drug target identification is a complex problem. Multiple evidence types must be integrated to understand disease biology and there are many gaps in our knowledge. The Open Targets research programme takes a systematic approach to two overarching but interconnected themes: first, to generate new data that associate targets with disease; and second, to integrate new and existing disease association evidence into a central informatic framework.

Open Targets initially hosted industry–academic workshops to understand the key questions asked in target discovery and the evidence required to address these questions. The focus was on integrating genetics, genomics and other relevant evidence to systematically assess target–disease associations.

Now, the core themes of the research programme are built through active discussion between all the academic and industry partners. Projects are developed within common areas of therapeutic and technological interest and are conducted collaboratively through the entire project lifecycle. To establish research projects, expressions of interest from both academia and industry are sought, then discussions held between groups with related interests to build project proposals for collective review. Each new round of project generation builds upon the interests and expertise of partners and the lessons learned from active projects.

Examples and evolution of projects

Open Targets projects systematically generate causal data that associate genes to disease or build foundational understanding of disease processes. Early projects harnessed the expertise and advanced technologies available at the Wellcome Sanger Institute, such as CRISPR gene perturbation and single-cell transcriptomics, to generate new data at scale. For example, an initial project involved conducting genome-wide CRISPR gene knockout screens across 324 cancer cell lines. This dataset contributed to the causal evidence supporting the DNA helicase WRN as a promising target in microsatellite-unstable colorectal cancers[2]. Another early project created foundational resources by using single-cell transcriptomics to generate atlases of airway and lung tissue from patients with asthma and those in remission.

Projects focussed on new data generation are centred around therapy areas of common interest to the consortium partners: oncology, immunology and neurodegeneration. This focus enables projects to learn from each other, share resources and take aligned approaches (for example, using the same induced pluripotent stem cell line background across projects). It also enables the collective analysis of multiple project outputs. Causal target–disease association studies in human cellular models of disease have expanded to more complex disease-relevant assays. For example, these assays investigate the impact of co-culturing immune cells and cancer cells, the role of multiple immune cell phenotypes in immune and neurodegenerative diseases, as well as aspects of neuronal function and the cellular stress response. With the emergence of new technologies and an expanded perturbation toolbox, the latest projects aim to generate hypotheses for precision therapeutics by integrating systematic combination screening in cancer, or base-editing in more complex systems, with single-cell readouts.

More recent foundational studies run by Open Targets increasingly focus on insights from patients to aid translation of findings to drug discovery decision-making. Projects in inflammatory bowel disease (IBD) and systemic lupus erythematosus use single-cell approaches to build detailed tissue maps in disease and generate cell-specific maps of quantitative trait loci to deconvolute genetic influences at cellular resolutions. A project in non-small-cell lung cancer profiles disease tissue using single-cell and spatial techniques, and another in metabolic dysfunction-associated steatohepatitis investigates disease progression, with both projects generating foundational datasets in diseases with high unmet need[3].

Datasets for assessing target–disease associations were initially identified within global, public data repositories, such as ChEMBL and UniProt hosted at EMBL-EBI. An ontology — a controlled vocabulary of diseases and phenotypes and the relationships between them — was required to map trait and disease terms across resources to bring the evidence into a common framework. Working closely with the EMBL-EBI Samples, Phenotypes and Ontologies Team, the Experimental Factor Ontology was selected due to its flexibility to incorporate terms and map across to other ontologies. A new informatics resource was then created to integrate and score the evidence from each different data type, providing overall a systematic assessment of target–disease associations based on evidence in a high-quality user interface: the Open Targets Platform[4].

New informatic projects build on the initial data pipelines that supported the Open Targets Platform, which required high degrees of manual oversight. Workflows have been refined and automated, while expanding coverage and knowledge extraction through more complex entity recognition, specific curation efforts and application of machine learning (ML). Open Targets has focused on a systematic approach to analysing genetic evidence for common traits and diseases, an approach supported by additional research demonstrating the impact of human genetic evidence on success rates in drug discovery. Collaborative projects with the NHGRI-EBI catalogue of human genome-wide association studies (GWAS Catalogue) enabled full summary statistics from GWAS to be available for Open Targets and the community. Open Targets then addressed the challenge of assigning causal genes to GWAS loci at scale by creating a new focussed resource: Open Targets Genetics. This companion to the Open Targets Platform provides large-scale genetic evidence on the relationships between disease-associated genetic loci and candidate causal variants, in order to predict drug target genes. The open-source resources provided by the Open Targets Platform have been adopted by academic and industry communities exploring target–disease associations. Significant efforts are underway to re-develop Open Targets Genetics pipelines to power future versions of the Open Targets informatics ecosystem, alongside applications of ML, to enable new modes of interacting with and making predictions from data.

Conclusion and outlook

The founding and successful development of Open Targets exemplifies effective pre-competitive collaboration between multiple commercial entities and academia, providing a model for consideration in other arenas. Feedback from Open Targets partners is that the most successful collaborations have maintained a strong, ongoing connection between academia and industry, driven by early engagement in project design and discussion of drug discovery questions throughout the project life cycle. The partners also highlight that the larger, more ambitious projects, applying new technology at scale, are of the highest scientific impact and enable multiple lines of analysis for drug discovery decision making through the richness of data generated. Building on that theme, Open Targets has recently launched Open-IBD, the consortium's most ambitious project to date, following 1,000 patients with IBD from diagnosis through to the early years of disease and treatment.

The research themes of Open Targets' programme have different impact profiles for drug target identification and prioritization depending on the nature of the findings. Causal target–disease associations, such as genetic perturbations that impact a disease-relevant phenotype, directly inform the decisions and investments of industry partners' pipeline of targets. Foundational knowledge, such as a disease tissue map or detailed characterization of a disease-relevant process, informs decision making less directly but more broadly, with such data being used many times across a class of targets or therapy area. Informatic infrastructure and resources are widely adopted to support overall decision making and target–disease research and provide sustainability for new knowledge generated. Open Targets' systematic approach produces lists of targets associated with disease, and further prioritization is required to build therapeutic hypotheses and for effective decision-making. Therefore, a later focus of the research programme has been target prioritization. Tools are being developed to assess the tractability, safety and feasibility of progressing potential drug targets, as well as to assess the novelty of targets and the potential therapeutic effect[5].

Looking ahead, Open Targets will continue to bring together academic and industry parties around ambitious projects to understand and treat disease. This approach should aid the translation of knowledge and technology from academic frontiers to industry applications and ultimately catalyse the development of safe and effective new therapies.

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Competing interests

G.T. consults for BMS, Genentech, GSK and Relation Tx, and serves on the scientific advisory board of Variant Bio.